



Name _____

Today's big idea

Two ideas today.

Operators let you do maths and ask complicated questions. `(score) + 10` · `(score) > 10` · `<this>` and `<that>` · `join [Level ] (level)`

My Blocks let you take a group of blocks you keep rewriting, give it a name, and use it as one single block from then on.

"How do you make a cup of tea?" — seven steps. Draw a box around them and call it **MAKE TEA**. Now "make tea for four people" is `repeat 4 [MAKE TEA]` instead of 28 steps.

That's called **abstraction**, and it's one of the most powerful ideas in all of programming.

Words to know

Word	What it means
operator	A block that does maths or asks a question
My Block	A block you invent yourself, made of other blocks
define	The hat block where you say what your new block does
abstraction	Naming a group of steps so you can forget the details

Dragging tip: building `<<(score) > 10>` and `<(lives) > 1>` inside a tiny slot is horrible. Build the whole thing in an **empty space** first, then drag the finished assembly into the `if`.

Class activity A – Quiz Sprite

ask [What is 7 x 6?] and wait

SENSING

if `<(answer) = 42>` then

CONTROL

say [Correct!] for 2 seconds

LOOKS

change score by 1

VARIABLES

```
else
```

```
say (join [No, it was ] [42]) for 2 seconds
```

Add two more questions. Notice how much you're copying and pasting. Activity B fixes that.

Class activity B – Make Your Own Block

```
define celebrate
```

```
say [Correct!] for 1 second
```

```
change size by 25
```

```
start sound [Cheer]
```

```
wait 0.5 seconds
```

```
change size by -25
```

Now replace all three copies of your celebration with one `celebrate` block. **Change the definition once → all three change.** That's the payoff.

Your project – it gets harder as you get better

- ☐ A `speed` variable. Replace the fixed `wait 0.02` with `wait (speed) seconds`
- ☐ Every 10 points: level up and multiply `speed` by 0.85

```
forever
```

```
if <(score) > ((level) * 10)> then
```

```
  change level by 1
```

```
  set speed to ((speed) * 0.85)
```

```
  say (join [Level ] (level)) for 1 seconds
```

```
wait 0.1 seconds
```

- ☐ A rare **bonus item** worth 5 points that only shows up sometimes:

```
if <(pick random 1 to 5) = 1> then
```

```
  show
```

else

CONTROL

hide

LOOKS

- ☐ Make a `reset object` My Block and use it on every falling sprite
- ☐ Save it

Homework

1. **A rare treasure.** If you didn't finish the bonus item, do it now.
2. **Tidy up.** Find code you've written more than once and turn it into a My Block.
3. **Think about the showcase.** Next week your family comes to see your game. Have two sentences ready: *what your game is*, and *the hardest bug you fixed*.

Bonus challenges

1. A power-up that makes your player wider for five seconds, then back to normal.
2. Use `join` to show "Level 3 — Score 42" as one readout.
3. Use `and` or `or` for a combo condition — a bonus that only counts on a high level.
4. **Really hard:** replace your duplicated falling sprites with **clones**. Look up `create clone of myself`, `when I start as a clone`, `delete this clone`. Nobody is expected to get this one.

If something breaks

- **Game freezes on level up?** `speed` hit zero or went negative. Put a floor on it.
- **Level goes up hundreds of times?** The `if` keeps firing. Make sure `level` is in the condition.
- **Can't drop an operator in?** Build it in empty space first, then drag it in.
- **My Block missing on another sprite?** My Blocks belong to one sprite only.

Stuck? Put your hand up — that is what we are here for. And always save before you close the lid.